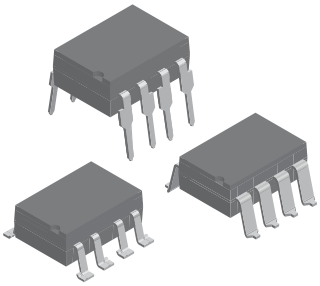
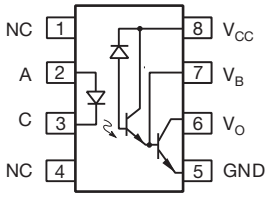
High Speed Optocoupler, 100 kBd, Low Input Current, Photodiode Darlington Output





FEATURES

- High current transfer ratio, 300 %
- Low input current, 0.5 mA
- High output current, 60 mA
- Isolation test voltage, 5300 V_{RMS}
- TTL compatible output, V_{OL} = 0.1 V
- High common mode rejection, 500 V/µs
- Adjustable bandwidth-access to base
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

Pb-free

e3

RoHS COMPLIANT

DESCRIPTION

High common mode transient immunity and very high current ratio together with 5300 V_{RMS} insulation are achieved by coupling and LED with an integrated high gain photo detector in an eight pin dual-in-line package. Separate pins for the photo diode and output stage enable TTL compatible saturation voltages with high speed operation.

Access to the base terminal allows adjustment to the gain bandwidth.

The 6N138 is ideal for TTL applications since the 300 % minimum current transfer ratio with an LED current of 1.6 mA enables operation with one unit load-in and one unit load-out with a 2.2 kΩ pull-up resistor.

The 6N139 is best suited for low power logic applications involving CMOS and low power TTL. A 400 % current transfer ratio with only 0.5 mA of LED current is guaranteed from 0 °C to 70 °C.

Caution: Due to the small geometries of this device, it should be handled with Electrostatic Discharge (ESD) precautions. Proper grounding would prevent damage further and/or degradation which may be induced by ESD.

APPLICATIONS

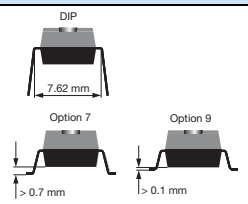
- Microprocessor system interface
- PLC, ATE input / output isolation
- EIA RS232 line receiver
- TTL, CMOS voltage level translation
- Multiplexed data transmission
- Digital control power supply
- Ground loop and electrical noise elimination

AGENCY APPROVALS

- UL1577, file no. E52744, double protection
- DIN EN 60747-5-5 available with option 1

ORDERING INFORMATION

6	N	1	3	#	-	X	0	#	#	T
PART NUMBER						PACKAGE OPTION				TAPE AND REEL



AGENCY CERTIFIED/PACKAGE	CTR (%)	
	1.6 mA	
UL	**> 300**	**> 500**
DIP-8	6N138	6N139, 6N139-X001
SMD-8, option 7	6N138-X007T	6N139-X007, 6N139-X007T
SMD-8, option 9	6N138-X009T	6N139-X009, 6N139-X009T
VDE	**> 300**	**> 500**
SMD-8, option 7	-	6N139-X017T
SMD-8, option 9	-	6N139-X019T

Note

- For additional information on the available options refer to option information.

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ABSOLUTE MAXIMUM RATINGS (T_{amb} = 25 °C, unless otherwise specified)					
PARAMETER	**TEST CONDITION**	**PART**	**SYMBOL**	**VALUE**	**UNIT**
INPUT					
Reverse voltage			V_R	5	V
Forward current			I_F	25	mA
Average input current (1)			$I_{f(avg)}$	20	mA
Input power dissipation (2)			P_{diss}	35	mW
OUTPUT					
Supply and output voltage	Pin 8 to 5, pin 6 to 5	6N138	V_{CC}, V_O	-0.5 to 7	V
	Pin 8 to 5, pin 6 to 5	6N139	V_{CC}, V_O	-0.5 to 18	V
Emitter base reverse voltage	Pin 5 to 7			0.5	V
Peak input current	50 % duty cycle - 1 ms pulse width			40	mA
Peak transient input current	$t_p \leq 1$ µs, 300 pps			1	A
Output current	Pin 6		I_O	60	mA
Output power dissipation (3)(4)			P_{diss}	100	mW
COUPLER					
Isolation test voltage	t = 1 min		V_{ISO}	5300	V_{RMS}
Isolation resistance	V_{IO} = 500 V, T_{amb} = 25 ° C		R_{IO}	$\geq 10^{12}$	Ω
	V_{IO} = 500 V, T_{amb} = 100 ° C		R_{IO}	$\geq 10^{11}$	Ω
Storage temperature			T_{stg}	-55 to +150	°C
Operating temperature			T_{amb}	-55 to +100	°C
Lead soldering temperature (5)	t = 10 s		T_{sld}	260	°C

Notes

- Stresses in excess of the absolute maximum ratings can cause permanent damage to the device. Functional operation of the device is not implied at these or any other conditions in excess of those given in the operational sections of this document. Exposure to absolute maximum ratings for extended periods of the time can adversely affect reliability.

(1) Derate linearly above 50 °C free-air temperature at a rate of 0.333 mA/°C.
(2) Derate linearly above 50 °C free-air temperature at a rate of 0.467 mW/°C.
(3) Derate linearly above 25 °C free-air temperature at a rate of 0.6 mA/°C.
(4) Derate linearly above 25 °C free-air temperature at a rate of 1 mW/°C.
(5) Refer to reflow profile for soldering conditions for surface mounted devices (SMD). Refer to wave profile for soldering conditions for through hole devices (DIP).

ELECTRICAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)							
PARAMETER	**TEST CONDITION**	**PART**	**SYMBOL**	**MIN.**	**TYP.**	**MAX.**	**UNIT**
INPUT							
Input forward voltage	I_F = 1.6 mA		V_F		1.4	1.7	V
Input reverse breakdown voltage	I_R = 10 µA		B_{VR}	5			V
Temperature coefficient of forward voltage	I_F = 1.6 mA				-1.8		mV/°C
OUTPUT							
Logic low, output voltage (1)	I_F = 1.6 mA, I_O = 4.8 mA, V_{CC} = 4.5 V	6N138	V_{OL}		0.1	0.4	V
	I_F = 1.6 mA, I_O = 8 mA, V_{CC} = 4.5 V	6N139	V_{OL}		0.1	0.4	V
	I_F = 5 mA, I_O = 15 mA, V_{CC} = 4.5 V	6N139	V_{OL}		0.15	0.4	V
	I_F = 12 mA, I_O = 24 mA, V_{CC} = 4.5 V	6N139	V_{OL}		0.25	0.4	V
Logic high, output current (1)	I_F = 0 mA, V_{CC} = 7 V	6N138	I_{OH}		0.1	250	µA
	I_F = 0 mA, V_{CC} = 18 V	6N139	I_{OH}		0.05	100	µA
Logic low supply current (1)	I_F = 1.6 mA, V_O = OPEN, V_{CC} = 18 V		I_{CCL}		0.2	1.5	mA
Logic high supply current (1)	I_F = 0 mA, V_O = OPEN, V_{CC} = 18 V		I_{CCH}		0.001	10	µA

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ELECTRICAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)							
PARAMETER	**TEST CONDITION**	**PART**	**SYMBOL**	**MIN.**	**TYP.**	**MAX.**	**UNIT**
COUPLER							
Input capacitance	f = 1 MHz, V_F = 0		C_{IN}		25		pF
Input output insulation leakage current (2)	45 % relative humidity, T_{amb} = 25 °C, t = 5 s, V_{IO} = 3000 V_{DC}					1	µA
Resistance (input to output) (2)	V = 500 V_{peak}		R_{IO}		10^{12}		Ω
Capacitance (input to output) (2)	f = 1 MHz		C_{IO}		0.6		pF

Notes

- Minimum and maximum values are testing requirements. Typical values are characteristics of the device and are the result of engineering evaluation. Typical values are for information only and are not part of the testing requirements.

(1) Pin 7 open.

(2) Device considered a two-terminal device: pins 1, 2, 3 and 4 shorted together and pins 5, 6, 7, and 8 shorted together.

CURRENT TRANSFER RATIO							
PARAMETER	**TEST CONDITION**	**PART**	**SYMBOL**	**MIN.**	**TYP.**	**MAX.**	**UNIT**
Current transfer ratio (1)(2)	I_F = 1.6 mA, V_O = 0.4 V, V_{CC} = 4.5 V	6N138	CTR	300	1600		%
Current transfer ratio	I_F = 0.5 mA, V_O = 0.4 V, V_{CC} = 4.5 V	6N139	CTR	400	1600		%
	I_F = 1.6 mA, V_O = 0.4 V, V_{CC} = 4.5 V	6N139	CTR	500	2000		%

Notes

(1) DC current transfer ratio is defined as the ratio of output collector current, I_O, to the forward LED input current, I_F times 100 %.

(2) Pin 7 open.

SAFETY AND INSULATION RATINGS						
PARAMETER	**TEST CONDITION**	**SYMBOL**	**MIN.**	**TYP.**	**MAX.**	**UNIT**
Climatic classification	according to IEC 68 part 1			55/100/21		
Comparative tracking index		CTI	175		399	
V_{IOTM}			8000			V
V_{IORM}			890			V
P_{SO}					500	mW
I_{SI}					300	mA
T_{SI}					175	°C
Creepage distance	Standard DIP-8		8			mm
Clearance distance	Standard DIP-8		7			mm
Creepage distance	400 mil DIP-8		8			mm
Clearance distance	400 mil DIP-8		8			mm

Note

- As per IEC 60747-5-2, §7.4.3.8.1, this optocoupler is suitable for “safe electrical insulation” only within the safety ratings. Compliance with the safety ratings shall be ensured by means of protective circuits.

SWITCHING CHARACTERISTICS							
PARAMETER	**TEST CONDITION**	**PART**	**SYMBOL**	**MIN.**	**TYP.**	**MAX.**	**UNIT**
Propagation delay time to logic low at output	I_F = 1.6 mA, R_L = 2.2 kΩ	6N138	t_{PHL}		2	10	µs
Propagation delay time to logic low at output (1)(2)	I_F = 0.5 mA, R_L = 4.7 kΩ	6N139	t_{PHL}		6	25	µs
	I_F = 12 mA, R_L = 270 Ω	6N139	t_{PHL}		0.6	1	µs
Propagation delay time to logic high at output	I_F = 1.6 mA, R_L = 2.2 kΩ	6N138	t_{PLH}		2	35	µs
	I_F = 0.5 mA, R_L = 4.7 kΩ	6N139	t_{PLH}		4	60	µs
Propagation delay time to logic high at output (1)	I_F = 12 mA, R_L = 270 Ω	6N139	t_{PLH}		1.5	7	µs

Note

(1) Using a resistor between pin 5 and 7 will decrease gain and delay time.

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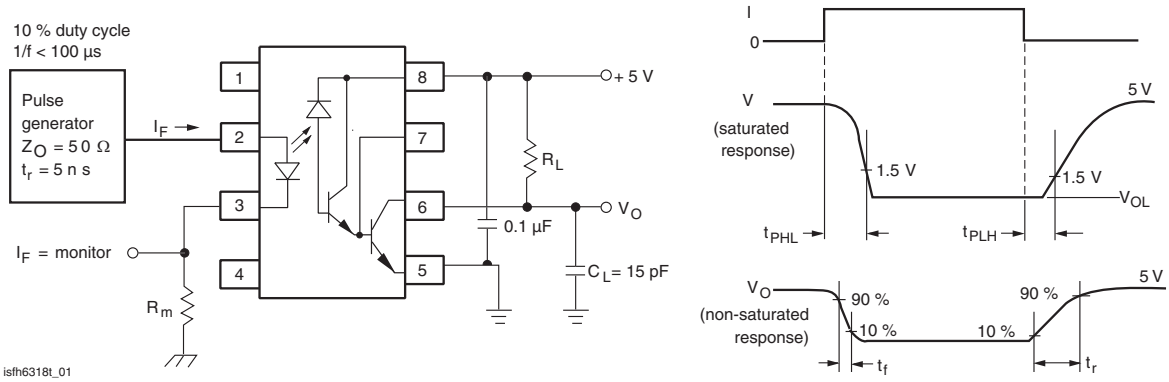


Fig. 1 - Switching Test Circuit

COMMON MODE TRANSIENT IMMUNITY							
PARAMETER	**TEST CONDITION**	**PART**	**SYMBOL**	**MIN.**	**TYP.**	**MAX.**	**UNIT**
Common mode transient immunity, logic high level output (1)(2)	I_F = 0 mA, R_L = 2.2 kΩ, R_{CC} = 0, $\|V_{CM}\|$ = 10 V_{P-P}		$\|CM_H\|$		500		V/µs
Common mode transient immunity, logic low level output (1)(2)	I_F = 16 mA, R_L = 2.2 kΩ, R_{CC} = 0, $\|V_{CM}\|$ = 10 V_{P-P}		$\|CM_L\|$		- 500		V/µs

Notes

(1) Common mode transient immunity in logic high level is the maxium tolerable (positive) dVcm/dt on the leading edge of the common mode pulse, V_{CM}, to assure that the output will remain in a logic high state (i.e. V_O > 2 V) common mode transient immunity in logic low level is the maximum tolerable (negative) dVcm/dt on the trailing edge of the common mode pulse signal, V_{CM} to assure that the output will remain in a logic low state (i.e. V_O < 0.8 V).

(2) In applications where dV/dt may exceed 50 000 V/µs (such as state discharge) a series resistor, R_{CC} should be included to protect I_C from destructively high surge currents. The recommend value is $R_{CC} \cong [(1\ V)/(0.15\ I_F\ (mA)]\ k\Omega$

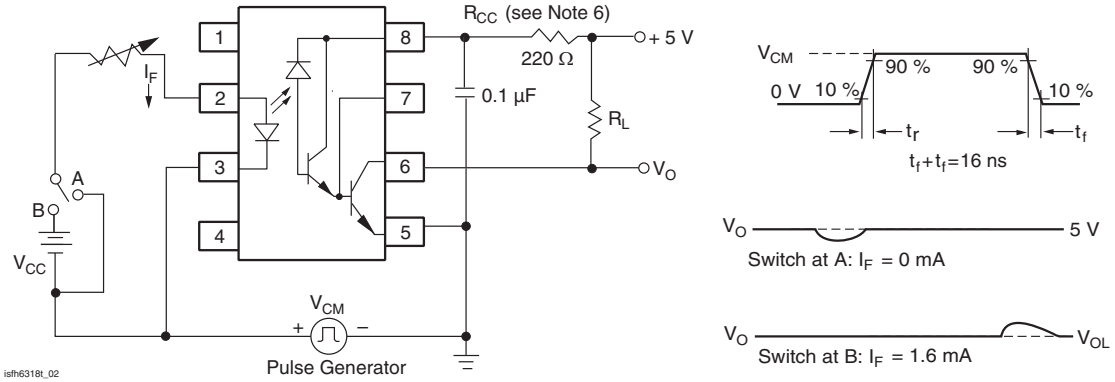


Fig. 2 - Test Circuit for Transient Immunity and Typical Waveforms

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TYPICAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)

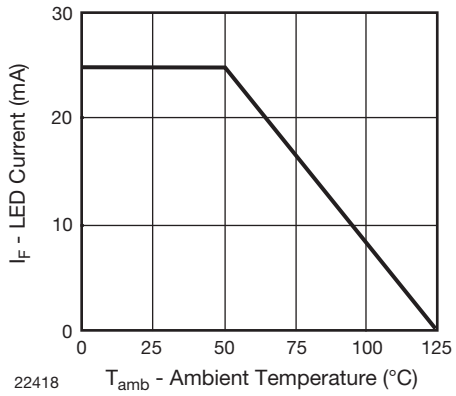


Fig. 3 - Permissible Forward LED Current vs. Temperature

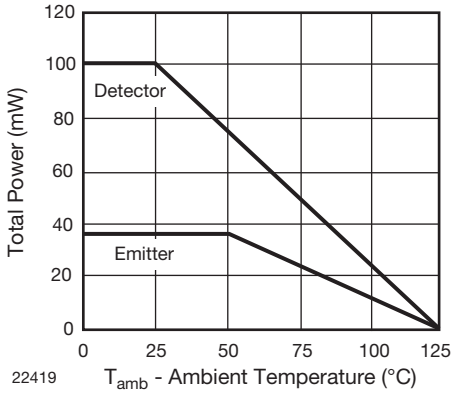
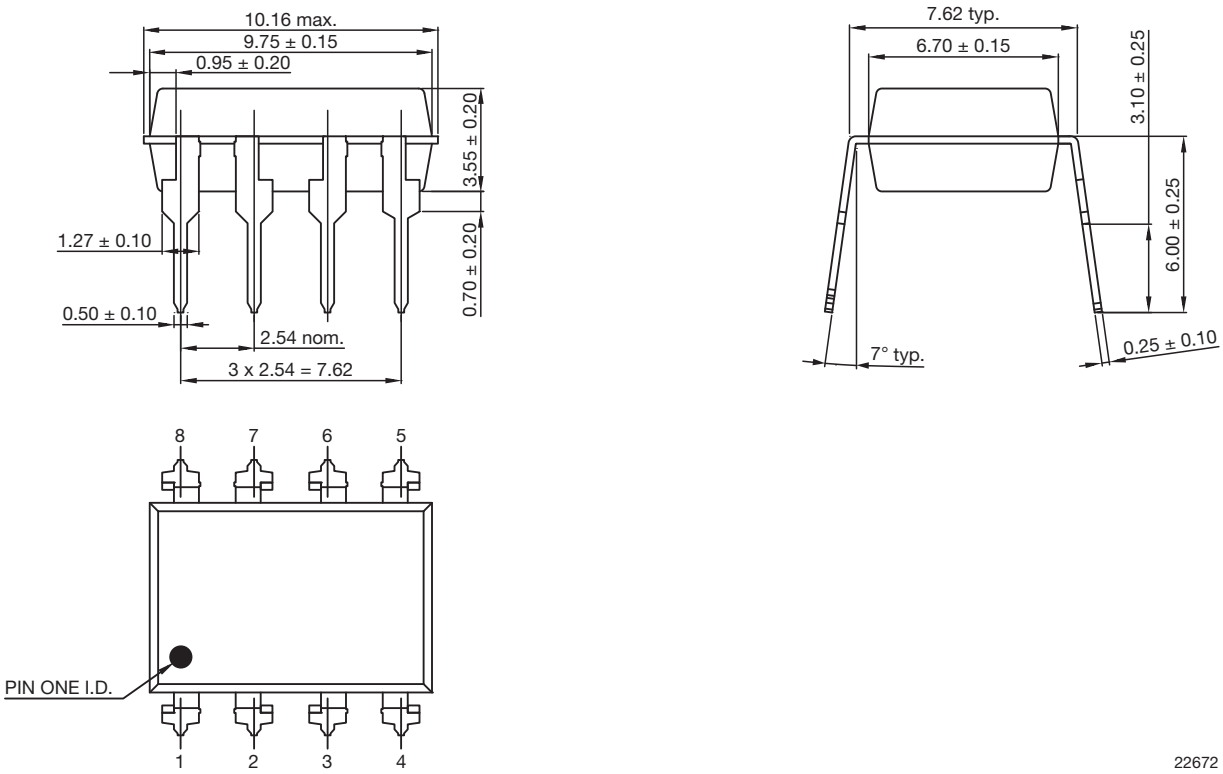


Fig. 4 - Permissible Power Dissipation vs. Temperature

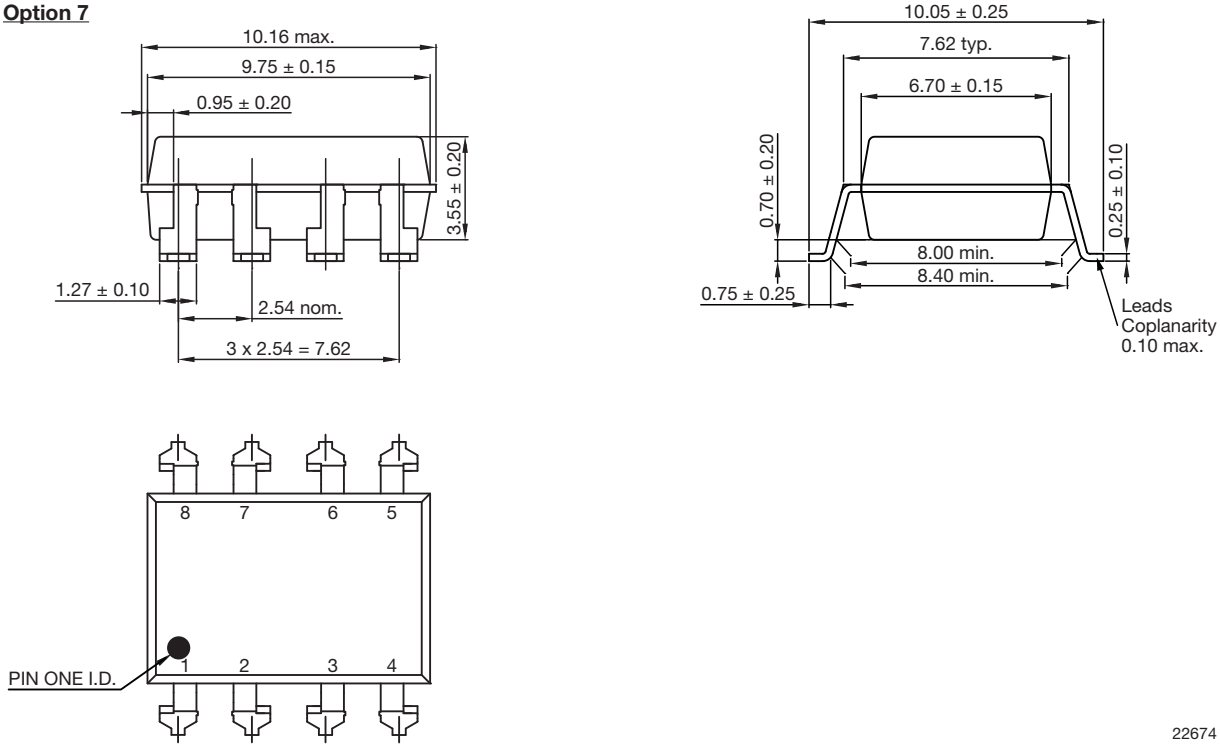
PACKAGE DIMENSIONS in millimeters

Standard

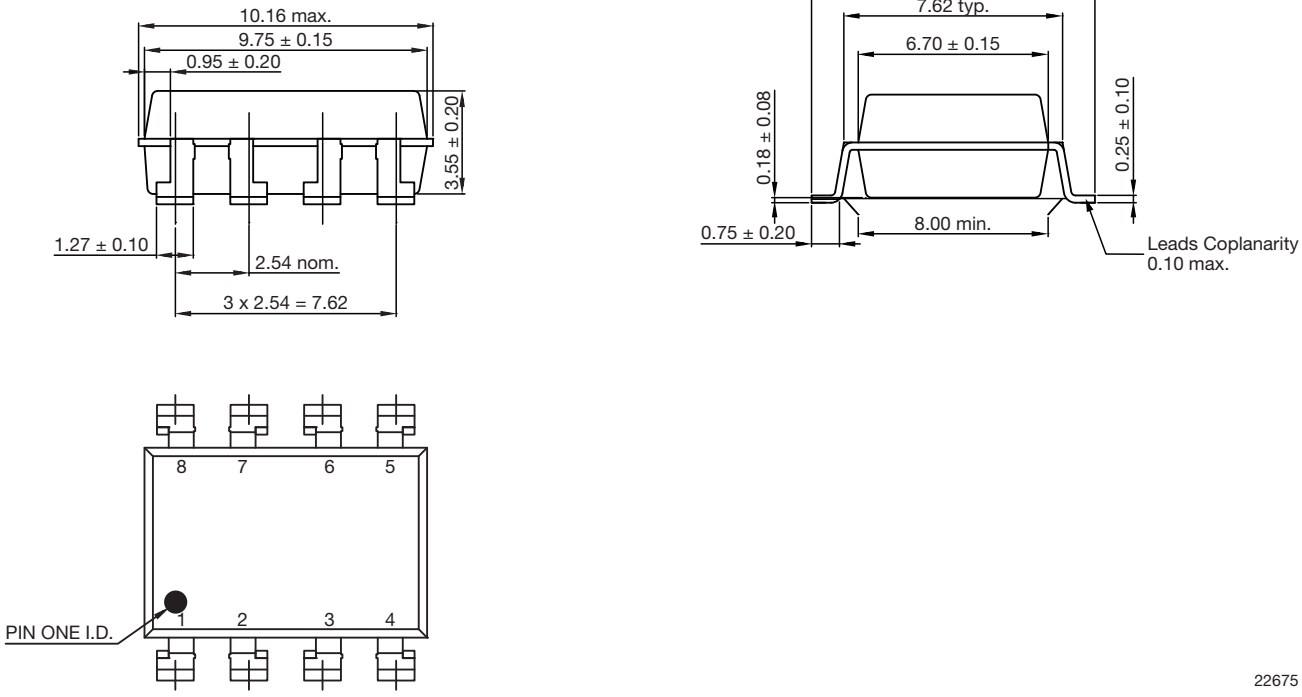


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Option 7

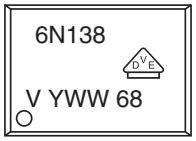


Option 9



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PACKAGE MARKING (example)



Note

- Basic product marking only, refer to “Option Information” document (www.vishay.com/doc?83713) for option marking

TAPE AND REEL DIMENSIONS in millimeters

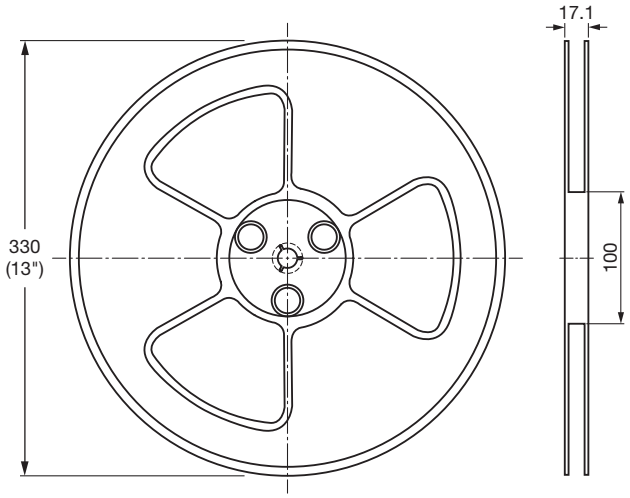


Fig. 3 - Reel Dimensions (3000 units per reel)

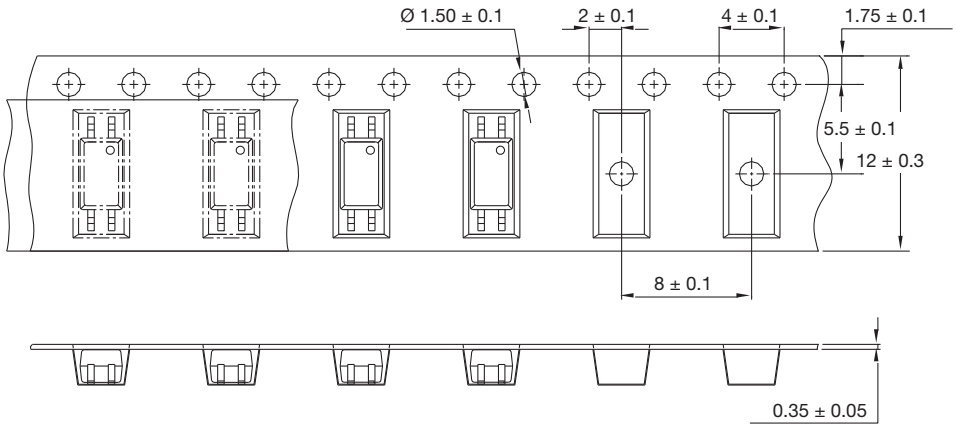


Fig. 4 - Tape Dimensions

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SOLDER PROFILES

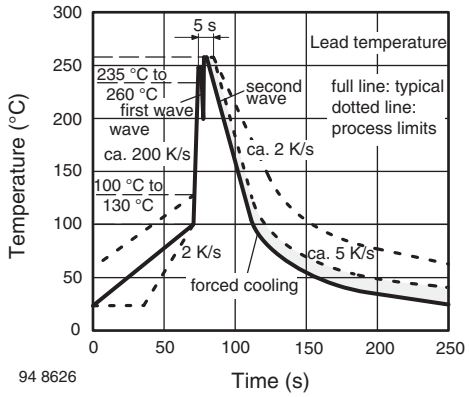


Fig. 5 - Wave Soldering Double Wave Profile According to J.STD-020 for DIP-8 Devices

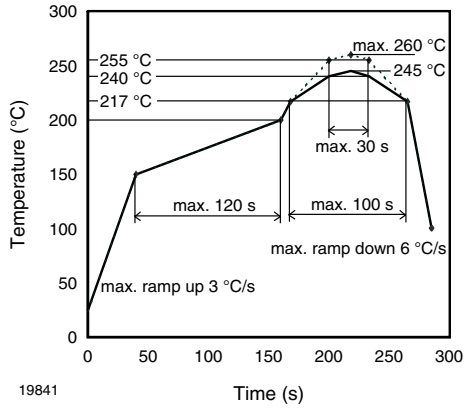


Fig. 6 - Lead (Pb)-free Reflow Solder Profile According to J-STD-020 for SMD-8 Devices

HANDLING AND STORAGE CONDITIONS

ESD level: HBM class 2

Floor life: unlimited

Conditions: T_{amb} < 30 °C, RH < 85 %

Moisture sensitivity level 1, according to J-STD-020

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